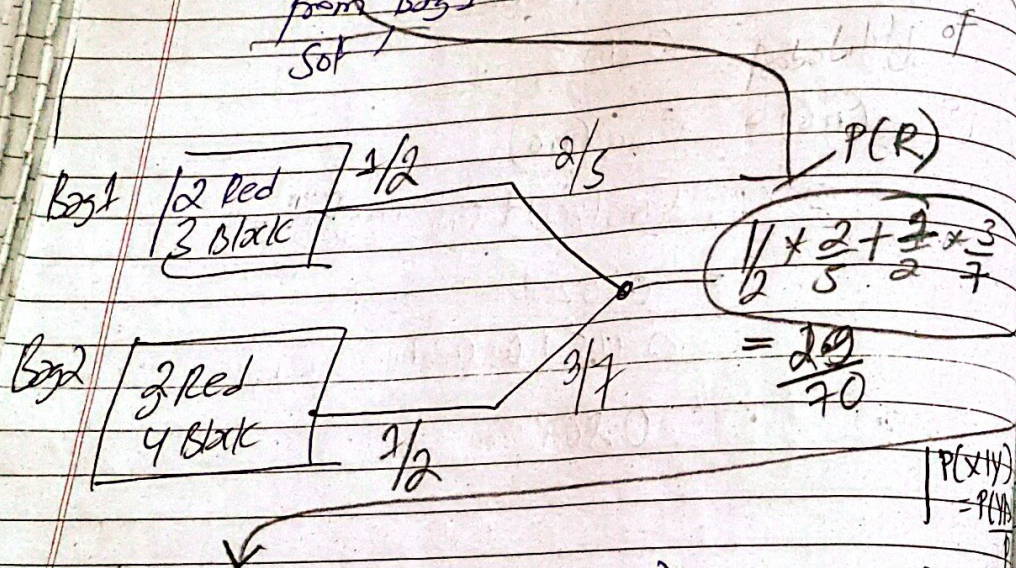


Another

7. Bag 1 contain 2 Red & 3 Black balls. Bag 2 contain 3 Red & 4 Black balls - One ball is drawn at random from one of these bags & is red. find the probability it is drawn from Bag 1.

Sol



$$P(B_1|R) = \frac{P(R|B_1) * P(B_1)}{P(R)}$$

$$= \frac{2/5 * 1/2}{29/70} = 0.4$$

$$\frac{29}{70}$$

$$P(Y|X) = \frac{P(X|Y) * P(Y)}{P(X)}$$

dataset = Y

features

$$P(Y|X_1, X_2, \dots, X_N)$$

(Assuming, every feature OR independent of each other)

$$= \frac{P(X_1|Y) * P(X_2|Y) * \dots * P(X_N|Y) * P(Y)}{P(X_1) * P(X_2) * \dots * P(X_N)} \quad (I)$$

dataset = NO

features

$$P(N|X_1, X_2, \dots, X_N)$$

$$= \frac{P(X_1|N) * P(X_2|N) * \dots * P(X_N|N) * P(N)}{P(X_1) * P(X_2) * \dots * P(X_N)} \quad (II)$$

Imp

$$\frac{1}{(II) + (I)} \approx \frac{1}{II}$$

So, we can neglect the denominator & only calculate numerator

$$P(Y|X_1, X_2, \dots, X_N) = P(X_1|Y) * P(X_2|Y) * \dots * P(X_N|Y) * P(Y)$$

Person	x_1 COVID (Y/N)	x_2 Flu (Yes/No)	Fever (Yes/No)
1	Yes	No	Yes
2	No	Yes	Yes
3	Yes	Yes	Yes
4	No	No	No
5	Yes	No	Yes
6	No	No	Yes
7	Yes	No	Yes
8	Yes	No	No
9	No	Yes	Yes
10	No	Yes	No

Ques, Person (flu, covid)

If person has both flu & covid then fever will be?

Ques

$$P(\text{fever} = \text{yes} | \text{flu, covid} = \text{yes}) = P(\text{flu} | \text{fever} = \text{yes}) * P(\text{covid} | \text{fever} = \text{yes})$$

Ans

$$P(\text{fever} = \text{No} | \text{flu, covid} = \text{yes}) = P(\text{flu} | \text{fever} = \text{No}) * P(\text{covid} | \text{fever} = \text{No})$$

So, probability

Eq (1)

$$P(\text{fever} = \text{yes} | \text{flu, covid}) = \frac{3}{9} * \frac{4}{7} * \frac{7}{10} = 0.121$$

$$P(\text{fever} = \text{No} | \text{flu, covid}) = \frac{1}{3} * \frac{1}{3} * \frac{3}{10}$$

Step 1 - Prior probability:-

$$P(\text{fever} = \text{yes}) = \frac{7}{10}$$

$$P(\text{fever} = \text{No}) = \frac{3}{10}$$

Step 2 - Conditional probability:- fever

cond	yes	No
yes	5/7	1/3
No	5	3/7

flu	yes	No
yes	4	3/7
No	6	4/7

PCA

principal component

VS

manifold

fever → yes (7)
→ No (3)

Pipeline

Hyperparameter

Tuning

Grid Search

Date

Page

Normalized probabilities

Date _____
Page _____

for actual probability values

$$P(\text{fever} = \text{yes} | \text{flu, cold}) = \frac{0.171}{0.171 + 0.033} = 0.839$$

$$P(\text{fever} = \text{No} | \text{flu, cold}) = \frac{0.033}{0.171 + 0.033} = 0.162$$

So, he'll have fever.